INCH-POUND
A-A-52491A
18 April 2018
SUPERSEDING
A-A-52491
April 15, 1994

COMMERCIAL ITEM DESCRIPTION

BEARING, GROOVED,
BALL, THRUST: INCH AND METRIC SIZES, RACE SURFACE RIGID OR FLAT SEAT TYPE

The General Services Administration has authorized the use of this commercial item description (CID), for all federal agencies.

1. SCOPE. This CID covers the requirements for a thrust ball bearing, referred to herein as "bearing". It consists of two hardened and ground washers (thrust plates) enclosing a complement of precision ground and lapped balls and. separated by a cage or retainer. The bearing is used to provide axial rigidity and not recommended if radial loading is expected.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein or in the applicable engineering drawing as cited in the contract, the materials used shall be in accordance with the manufacturer's material specifications/drawings.

2.2 Thrust plates and balls. The material for thrust plates and balls shall be vacuum degassed bearing quality steel.

2.3 Heat treatment. The metal parts of the bearing shall be through hardened to Rockwell C-58 to C-66.

2.2 Design and construction. The general design and construction features of the thrust bearing are shown in Figure 1. The PIN number of a given size thrust bearing is shown in Table I. Detailed and specific features and be as specified in the contract.

2.2.1 Size. The size of the bearing shall be determined based on specific application requirements and shall be specified in the contract. Tolerances are given in Table I for preferred sizes. Factors such as load, rpm, environmental conditions (e.g., temperature, humidity) are taken into consideration. The basic empirical formula for bearing load is explained in 6.2.

2.2.2 Radius or chamfer. Bearings must clear the maximum fillet radius of mating shaft or housing. A radius or chamfer (see Figure 1), 0.031 inch minimum, shall be used on the bore or outer diameter.

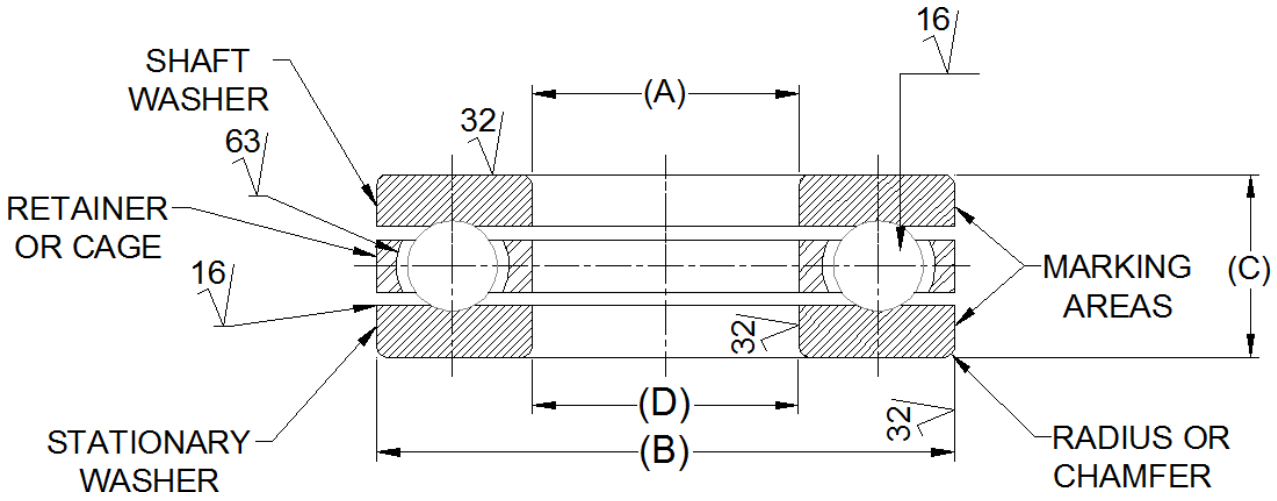
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2.2.3 Surfaces. Surfaces marked √ in Figure 1 shall represent maximum surface roughness in microinch (µin). The ground surfaces shall be free from tool marks, chatter waves, pits, rust, or other injurious defects. Other surfaces shall have a smooth finish, free from scale. The bearings shall be supplied without lubrication, but with a coating of rust preventing film.



(A) = Optimal inside (bore) diameter
(B) = Optimal outside diameter
(C) = Optimal height
(D) = Optimal large bore diameter, when specified in contract.

2.3 Identification markings. Identification markings shall be of a size and location easily readable with normal vision at a distance of 12 inches, on outermost surface (see Figure 1 – marking areas). Marking shall be permanent, legible and shall include, as a minimum, the manufacturer's CAGE code, (the PIN) A52491- (and the appropriate dash number (size code), from Table I (see Table I).

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TABLE I. PIN numbers with preferred sizes/tolerances of thrust bearings[1]

PIN dash numbers below	DIAMETERS AND TOLERENCE					OUTSIDE DIAMETER		HEIGHT			RADIUS MAXIMUM	BASIC LOAD DYNAMIC RATING (POUNDS)	BASIC LOAD STATIC RATING (POUNDS)
	DIAMETER (A)		TOLERANCE (A & D)	DIAMETER (D)		(B) TOLERENCE MM +0.0000 -0.0500 INCH +0.0000 -0.0020		(C)		TOLERENCE ±			
	MM	INCH		MM	INCH	MM	INCH	MM	INCH				
1		0.3750	-0.0000 +0.0008		0.4063		1.0000		0.5313	0.005	0.0300	1220	1500
2	10	0.3937	-0.0004 +0.0004	10.2	0.4016	30	1.1811	12	0.4724	0.005	0.0200	1960	2720
3	10	0.3937	-0.0004 +0.0004	10.2	0.4016	26	1.0236	12	0.4724	0.005	0.0200	1910	2720
4	12	0.4724	-0.0004 +0.0004	12.2	0.4803	28	1.1024	12	0.4724	0.005	0.0200	1650	2500
5		0.5000	-0.0040 +0.0060		0.5000		1.2188		0.5626	0.005	0.0300	1810	3000
6		0.5000	-0.0000 +0.0008		0.5313		1.2813		0.6250	0.005	0.0300	2370	3500
7		0.5625	-0.0040 +0.0060		0.5625		1.2188		0.5625	0.005	0.0300	1790	3000
8	15	0.5906	-0.0004 +0.0004	15.2	0.5985	31	1.2205	12	0.4724	0.005	0.0200	1780	3000
9	15	0.5906	-0.0004 +0.0004	15.2	0.5985	35	1.3780	14	0.5512	0.005	0.0200	2260	3550
10		0.6250	-0.0040 +0.0060		0.6250		1.3438		0.5625	0.005	0.0300	1640	2750
11		0.6250	-0.0000 +0.0008		0.6563		1.4063		0.6250	0.005	0.0300	2250	3550
12	18	0.7087	-0.0004 +0.0004	18.2	0.7166	35	1.3780	12	0.4724	0.005	0.0200	1710	3000
13		0.7500	-0.0040 +0.0060		0.7500		1.4688		0.5625	0.005	0.0300	1860	3500
14		0.7500	-0.0000 +0.0008		0.7813		1.5313		0.6250	0.005	0.0300	2550	4440
15	20	0.7874	-0.0004 +0.0004	20.2	0.7953	37	1.4567	12	0.4724	0.005	0.0200	1670	3000
16	20	0.7874	-0.0004 +0.0004	20.2	0.7953	40	1.5748	14	0.5512	0.005	0.0200	2790	4880
17		0.8125	-0.0040 +0.0060		0.5125		1.4688		0.5625	0.005	0.0300	1840	3500
18	22	0.8661	-0.0004 +0.0004	22.2	0.8740	42	1.6535	14	0.5512	0.005	0.0200	2790	5320
19		0.8750	-0.0000 +0.0008		0.9063		1.6563		0.6250	0.005	0.0300	2790	4880
20		0.8750	-0.0040 +0.0060		0.8750		1.8438		0.6250	0.005	0.0300	28880	5320
21		0.9375	-0.0040 +0.0060		0.9375		1.8438		0.6250	0.005	0.0300	2870	5320
22	25	0.9843	-0.0000 +0.0008	25.2	0.9922	45	1.7717	14	0.5512	0.005	0.0200	2880	5320
23	25	0.9843	-0.0004 +0.0004	25.2	0.9922	48	1.8898	15.5	0.6103	0.005	0.0200	2880	5770
24		1.0000	-0.0004 +0.0004		1.0313		1.7813		0.6250	0.005	0.0300	2870	5270
25		1.0000	-0.0050 +0.0070		1.0000		1.9688		0.6250	0.005	0.0300	2810	5320
26		1.0000	-0.0050 +0.0070		1.0000		2.0625		0.7500	0.005	0.0300	4200	7603
27		1.0625	-0.0050 +0.0070		1.0625		1.9688		0.6250	0.005	0.0300	2800	5320
28		1.1250	-0.0050 +0.0070		1.1250		2.1875		0.7500	0.005	0.0300	4100	7630
29		1.1250	-0.0000 +0.0008		1.1563		1.9063		0.6250	0.005	0.0300	2800	5320
30	30	1.1811	-0.0004 +0.0004	30.2	1.1890	50	1.9685	15	0.5512	0.005	0.0200	2920	5770
31	30	1.1811	-0.0004 +0.0004	30.2	1.1890	53	2.0866	15.5	0.6103	0.005	0.0200	4740	9710
32		1.1875	-0.0050 +0.0070		1.1875		2.0938		0.6250	0.005	0.0300	3020	6210

NOTE: [1] Location of letters in parenthesis () can be found on Figure 1.

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TABLE I. PIN numbers with preferred sizes/tolerances of thrust bearings - Continued

PIN dash numbers below	DIAMETERS AND TOLERENCE					OUTSIDE DIAMETER		HEIGHT			RADIUS MAXIMUM	BASIC LOAD DYNAMIC RATING	BASIC LOAD STATIC RATING
	DIAMETER (A)		TOLERANCE (A & D)	DIAMETER (D)		(B) TOLERENCE MM +0.0000 -0.0500 INCH +0.0000 -0.0020		(C)		TOLERENCE ±			
	MM	INCH		MM	INCH	MM	INCH	MM	INCH				
33		1.2500	-0.0050 +0.0070		1.250		2.3125		0.7500	0.005	0.0300	4480	9020
34		1.2500	-0.0000 +0.0008		1.2813		2.0938		0.7188	0.005	0.0300	4350	9000
35		1.2500	-0.0050 +0.0070		1.250		2.3438		0.6250	0.005	0.0300	3350	7540
36		1.3125	-0.0050 +0.0070		1.3125		2.3438		0.6250	0.005	0.0300	3340	7540
37		1.3750	-0.0050 +0.0070		1.3750		2.4688		0.6250	0.005	0.0300	3280	7540
38		1.3750	-0.0000 +0.0008		1.4063		2.2188		0.7188	0.005	0.0300	3350	7540
39	35	1.3780	-0.0005 +0.0005	35.2	1.3859	55	2.1654	16	0.6299	0.005	0.0200	3350	7920
40	35	1.3780	-0.0005 +0.0005	35.2	1.3859	62	2.4409	18	0.7087	0.005	0.0200	4370	9220
41		1.5000	-0.0050 +0.0070		1.5000		2.5625		0.7500	0.005	0.0400	4960	11100
42		1.5000	-0.0050 +0.0070		1.5000		2.5938		0.6250	0.005	0.0400	3340	7990
43		1.5000	-0.0000 +0.0010		1.5313		2.3438		0.7188	0.005	0.0400	4370	9710
44		1.5625	-0.0050 +0.0070		1.5625		2.5625		0.7500	0.005	0.0400	3340	7990
45		1.5625	-0.0050 +0.0070		1.5625		2.5938		0.6250	0.005	0.0400	3330	7990
46	40	1.5748	-0.0005 +0.0005	40.2	1.5827	64	2.5197	18	0.7087	0.005	0.0200	5010	12000
47	40	1.5748	-0.0005 +0.0005	42	1.6535	60	2.3622	13	0.5118	0.005	0.0240	4660	11100
48	40	1.5748	-0.0005 +0.0005	40.2	1.5827	60	2.3622	16	0.6299	0.005	0.0200	4660	11100
49		1.6250	-0.0050 +0.0070		1.6250		2.8125		0.8750	0.005	0.0400	6780	15000
50		1.6250	-0.0050 +0.0070		1.6250		2.9688		0.8125	0.005	0.0400	4770	11100
51		1.6250	-0.0000 +0.0010		1.6562		2.4688		0.7188	0.005	0.0400	3960	8990
52		1.6875	-0.0050 +0.0070		1.6875		2.8125		0.8750	0.005	0.0400	6750	15000
53		1.6875	-0.0050 +0.0070		1.6875		2.9688		0.8125	0.005	0.0400	4760	11100
54		1.7500	-0.0050 +0.0070		1.7500		2.9375		0.8750	0.005	0.0400	6370	14000
55		1.7500	-0.0000 +0.0010		1.7813		2.6875		0.7500	0.005	0.0400	4760	11800
56		1.7500	-0.0050 +0.0070		1.7500		3.0937		0.8125	0.005	0.0400	4690	11100
57	45	1.7717	-0.0005 +0.0005	45.2	1.7796	68	2.6772	16	0.6299	0.005	0.0200	4500	11200
58	45	1.7717	-0.0005 +0.0005	47	1.8500	65	2.5591	14	0.5512	0.005	0.0240	4500	11200
59	45	1.7717	-0.0005 +0.0005	45.2	1.7796	73	2.8740	22	0.8661	0.005	0.0200	7260	17000
60		1.1825	-0.0050 +0.0070		1.8125		2.9375		0.8750	0.005	0.0400	6950	15970
61		1.1825	-0.0050 +0.0070		1.8125		3.0937		0.8125	0.005	0.0400	4680	11100
62		1.8750	-0.0000 +0.0010		1.9063		2.8125		0.7500	0.005	0.0400	4460	11080
63		1.8750	-0.0050 +0.0070		1.8750		3.0625		0.8750	0.005	0.0400	5970	13000
64		1.8750	-0.0050 +0.0070		1.8750		3.2188		0.8125	0.005	0.0400	5370	13900
65		1.9375	-0.0050 +0.0070		1.9375		3.0625		0.8750	0.005	0.0400	5940	13000
66		1.9375	-0.0050 +0.0070		1.9375		3.2188		0.8125	0.005	0.0400	5350	13900
67	50	1.9685	-0.0005 +0.0005	50.2	1.9764	74	2.9134	18	0.7087	0.005	0.0200	4690	11100
68	50	1.9685	-0.0005 +0.0005	52	2.0470	70	2.7559	14	0.5512	0.005	0.0240	---	---
69	50	1.9685	-0.0005 +0.0005	50.2	1.9764	78	3.0709	22	0.8661	0.005	0.0200	6540	15000
70		2.0000	-0.0000 +0.0010		2.0313		2.9688		0.7500	0.005	0.0400	4660	11100
71		2.0000	-0.0060 +0.0080		2.0000		3.1875		0.8750	0.005	0.0400	6760	16000
72		2.3625	-0.0060 +0.0080		2.0000		3.3438		0.8125	0.005	0.0400	5290	13900
73		2.0625	-0.0060 +0.0080		2.0625		3.1875		0.8750	0.005	0.0600	5740	16000
74		2.0625	-0.0060 +0.0080		2.0625		3.3438		0.8125	0.005	0.0600	5280	13900
75		2.1250	-0.0060 +0.0080		2.1250		3.3125		0.8750	0.005	0.0600	7140	17970
76		2.125	-0.0060 +0.0080		2.1250		3.5938		0.8125	0.005	0.0600	7090	18000
77	55	2.1654	-0.0006 +0.0006	55.2	2.1733	88	3.4646	24.5	0.9646	0.005	0.0200	8580	21700

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PIN dash numbers below	DIAMETERS AND TOLERENCE					OUTSIDE DIAMETER		HEIGHT			RADIUS MAXIMUM	BASIC LOAD DYNAMIC RATING	BASIC LOAD STATIC RATING
	DIAMETER (A)		TOLERANCE (A & D)	DIAMETER (D)		(B) TOLERENCE MM +0.0000 -0.0500 INCH +0.0000 -0.0020		(C)		TOLERENCE ±			
	MM	INCH		MM	INCH	MM	INCH	MM	INCH				
78	55	2.1654	-0.0006 +0.0006	55.2	2.1733	78	3.0709	18	0.7087	0.005	0.0200	5220	14600
79		2.1875	-0.0060 +0.0080		2.1875		3.125		0.8750	0.005	0.0600	7140	17970
80		2.1875	-0.0060 +0.0080		2.1875		3.5938		0.8125	0.005	0.0600	7070	18000
81		2.2500	-0.0000 +0.0010		2.2813		3.3438		0.8750	0.005	0.0600	5230	13900
82		2.2500	-0.0060 +0.0080		2.2500		3.4375		0.8750	0.005	0.0600	6560	16000
83		2.2500	-0.0060 +0.0080		2.2500		3.7188		0.8125	0.005	0.0600	7000	18000
84		2.3125	-0.0060 +0.0080		2.3125		3.4375		0.8750	0.005	0.0600	6540	16000
85		2.3125	-0.0060 +0.0080		2.3125		3.7188		0.8125	0.005	0.0600	6980	18000
86	60	2.3622	-0.0006 +0.0006	60.2	2.3701	82	3.2284	18	0.7087	0.005	0.0200	5420	15300
87	60	2.3622	-0.0006 +0.0006	60.2	2.3701	90	3.5433	24.5	0.9646	0.005	0.0600	8830	23070
88		2.3750	-0.0060 +0.0080		2.3750		3.5625		0.8750	0.005	0.0600	9690	24500
89		2.3750	-0.0060 +0.0080		2.3750		3.8438		0.8125	0.005	0.0600	6920	18000
90		2.4375	-0.0060 +0.0080		2.4375		3.5625		0.8750	0.005	0.0600	9680	24500
91		2.4375	-0.0060 +0.0080		2.4375		3.8438		0.8125	0.005	0.0600	6690	18000
92		2.5000	-0.0000 +0.0012		2.5313		3.7188		1.0000	0.005	0.0600	8740	23070
93		2.5000	-0.0060 +0.0080		2.5000		3.9688		0.8125	0.005	0.0600	6840	18000
94		2.5000	-0.0060 +0.0080		2.5000		4.0000		1.1250	0.005	0.0600	9430	24500
95	65	2.5591	-0.0006 +0.0006	65.2	2.5670	100	3.9370	27	1.0630	0.005	0.0400	11600	28400
96	65	2.5591	-0.0006 +0.0006	65.2	2.5670	90	3.5433	20	0.7874	0.005	0.0200	6400	18900
97		2.5625	-0.0060 +0.0080		2.5625		3.9688		0.8125	0.005	0.0600	6820	18000
98		2.5625	-0.0060 +0.0080		2.5625		4.0000		1.1250	0.005	0.0600	9400	24500
99		2.6250	-0.0060 +0.0080		2.6250		4.1250		1.1250	0.005	0.0600	10870	26620
100		2.6250	-0.0060 +0.0080		2.6250		4.3438		1.0000	0.005	0.0600	9920	27200
101		2.6875	-0.0060 +0.0080		2.6875		4.1250		1.1250	0.005	0.0600	10870	26620
102		2.6875	-0.0060 +0.0080		2.6875		4.3438		1.0000	0.005	0.0600	9880	27200
103		2.7500	-0.0000 +0.0012		2.7813		4.0313		1.0000	0.005	0.0600	8600	21700
104		2.7500	-0.0060 +0.0080		2.7500		4.2500		1.1250	0.005	0.0600	11720	30180
105		2.7500	-0.0060 +0.0080		2.7500		4.1325		1.0000	0.005	0.0600	11300	28400
106		2.7500	-0.0060 +0.0080		2.7500		4.4688		1.0000	0.005	0.0600	11200	28400
107	70	2.7559	-0.0006 +0.0006	70.2	2.7638	103	4.0551	27	1.0630	0.005	0.0400	9630	25800
108	70	2.7559	-0.0006 +0.0006	70.2	2.7638	95	3.7402	20	0.7874	0.005	0.0200	7390	22000
109		2.8125	-0.0060 +0.0080		2.8125		4.2500		1.1250	0.005	0.0600	11720	30180
110		2.8125	-0.0060 +0.0080		2.8125		4.4688		1.0000	0.005	0.0600	11200	28400
111		2.8750	-0.0060 +0.0080		2.8750		4.3750		1.1250	0.005	0.0600	12470	33720
112		2.8750	-0.0060 +0.0080		2.8750		4.5938		1.0000	0.005	0.0600	12400	33700
113		2.9375	-0.0060 +0.0080		2.9375		4.3750		1.1250	0.005	0.0800	12470	33720
114		2.9375	-0.0060 +0.0080		2.9375		4.5938		1.0000	0.005	0.0800	12400	33700
115	75	2.9528	-0.0006 +0.0006	75.2	2.9607	110	4.3307	27	1.0630	0.005	0.0400	11410	31950
116	75	2.9528	-0.0006 +0.0006	75.2	2.9607	100	3.9370	20	0.7874	0.005	0.0200	7150	21800
117		3.0000	-0.0000 +0.0012		3.0313		4.3750		1.1250	0.005	0.0800	11410	32000
118		3.0000	-0.0060 +0.0080		3.0000		4.5000		1.1250	0.005	0.0800	9370	25800
119		3.0000	-0.0060 +0.0080		3.0000		4.7188		1.0000	0.005	0.0800	12700	35500
120	80	3.1496	-0.0006 +0.0006	80.2	3.1575	110	4.3307	22	0.8661	0.005	0.0200	7370	23490
121	80	3.1469	-0.0006 +0.0006	80.2	3.1575	115	4.5276	29	1.1417	0.005	0.0400	11180	38120
122	85	3.3465	-0.0008 +0.0008	85.2	3.544	115	4.5276	22	0.8661	0.010	0.0200	7450	24300

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TABLE I. PIN numbers with preferred sizes/tolerances of thrust bearings - Continued

PIN dash numbers below	DIAMETERS AND TOLERENCE					OUTSIDE DIAMETER		HEIGHT			RADIUS MAXIMUM	BASIC LOAD DYNAMIC RATING	BASIC LOAD STATIC RATING
	DIAMETER (A)		TOLERANCE (A & D)	DIAMETER (D)		(B) TOLERENCE MM +0.0000 -0.0500 INCH +0.0000 -0.0020		(C)		TOLERENCE ±			
	MM	INCH		MM	INCH	MM	INCH	MM	INCH				
123	85	3.3465	-0.0008 +0.0008	85.2	3.544	125	4.9213	30.5	1.2008	0.010	0.0400	16210	44400
124		3.5000	-0.0080 +0.0100		3.500		5.0000		1.125	0.010	0.0800	9640	28500
125		3.5000	-0.0000 +0.0012		3.5313		5.0625		1.2500	0.010	0.0800	14480	42680
126	90	3.5433	-0.0008 +0.0008	90.2	3.5512	120	4.7244	22	0.8661	0.010	0.0200	8840	29300
127	90	3.5433	-0.0008 +0.0008	90.2	3.5512	135	5.3150	30.5	1.2008	0.010	0.0400	17170	49900
128	95	3.7402	-0.0008 +0.0008	95.2	3.7481	130	5.1181	25	0.9843	0.010	0.0200	11170	39000
129	95	3.7402	-0.0008 +0.0008	95.2	3.7481	140	5.5118	32.5	1.2795	0.010	0.0400	19930	57000
130	100	3.9370	-0.0008 +0.0008	100.2	3.9449	135	5.3150	25	0.9843	0.010	0.0200	11890	40600
131	100	3.9370	-0.0008 +0.0008	100.2	3.9449	150	5.9055	32.5	1.2795	0.010	0.0400	20300	60400
132		4.0000	-0.0000 +0.0012		4.0313		5.8125		1.3750	0.010	0.0800	15350	49400
133		4.0000	-0.0080 +0.0100		4.0000		6.0000		1.3750	0.010	0.0800	18970	55470
134	105	4.1339	-0.0008 +0.0008	105.2	4.1418	140	5.5118	25	0.9843	0.010	0.0200	11990	42100
135	105	4.1339	-0.0008 +0.0008	105.2	4.1418	155	6.1024	40	1.5748	0.010	0.0400	23250	67900
136	110	4.3307	-0.0008 +0.0008	110.2	4.3386	145	5.7087	25	0.9843	0.010	0.0200	13290	46200
137		4.5000	-0.0080 +0.0100		4.5000		6.7500		1.6250	0.010	0.1200	26450	79880
138	115	4.5276	-0.0008 +0.0008	115.2	4.5355	150	5.9055	25	0.9843	0.010	0.0200	13450	47900
139	115		-0.0008 +0.0008	115.2	4.5355	165	6.4961	43	1.6929	0.010	0.0400	27900	84400
140	120	4.7244	-0.0008 +0.0008	120.2	4.7323	160	6.2992	27	1.0630	0.010	0.0200	14700	51500
141	125	4.9213	-0.0008 +0.0008	125.2	4.9292	165	6.4961	27	1.0630	0.010	0.0200	15190	56100
142	125	4.9213	-0.0010 +0.0010	125.2	4.9292	175	6.8898	46.5	1.8307	0.010	0.0400	26410	87900
143	140	5.5118	-0.0010 +0.0010	140.2	5.5197	200	7.8740	52	2.0472	0.010	0.0400	40760	127800

3. PRODUCT CONFORMANCE PROVISIONS.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this commercial item description and the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. REGULATORY REQUIREMENTS.

4.1 Use of recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PACKAGING.

5.1 Packaging. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. PIN to include the dash number(s) (for the specific size(s) to be made).

6.2 Bearing load formula. The equivalent thrust load rating (in pounds) for any combination of rotational speed (in rpm) and rating life (in hours) may be found from the following formula:

$$PP_{AA} = \frac{100CC}{(60\ NNNN)1/3}$$

Where:

P_A = Equivalent Thrust Load Rating
C = Basic Dynamic Load Rating (Pounds)
N = Rotational Speed - (rpm)
H = Rating Life - (Hours)

a. Equivalent thrust load is defined as a constant, central, purely axial load which, if applied to a bearing with rotating shaft washer and stationary housing washer, would give the same life as the bearing will attain under the actual conditions of load and rotation.

b. Rating life is defined as the number of hours at some constant speed that 90% of a group of apparently identical bearings will complete or exceed before the first evidence of fatigue develops.

c. Basic dynamic load rating is defined as a constant, central, thrust load which a group of apparently identical bearings can endure for a rating life of one million revolutions of one of the bearing washers.

6.3 Key words:

Ball
Bearing
Flat
Groove
Inch
Metric
Race
Rigid
Seat
Size
Surface
Thrust
Type

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Custodian:
Army – AT
Air Force – 99

Review Activities:
Army – AR, MI
Air Force – 187
DLA – GS, IS
Navy – MC

Preparing Activity:
Army - AT

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